

Smart Helmet with Alcohol Detection and Emergency Alert System for Post-Accident Assistance

S.Nagendramoorthy
Department of Computer Science and Engineering
University of Moratuwa
Moratuwa, Sri Lanka
nagendramoorthy.22@cse.mrt.ac.lk

S.J. Sooriyaarachchi
Department of Computer Science and Engineering
University of Moratuwa
Moratuwa, Sri Lanka
sulochanas@cse.mrt.ac.lk

Keywords— *Smart Helmet, Alcohol Sensor, Crash Detection, IoT Integration, GSM Alert*

I. INTRODUCTION

Motorcycle accidents continue to be a major cause of fatalities worldwide, and the risk is significantly increased when riders consume alcohol. Rapid post-accident response is crucial to minimize casualties and improve outcomes for injured riders. This research presents a Smart Helmet with Alcohol Detection and Accident Alert System, designed to monitor helmet usage, detect alcohol consumption, and automatically send emergency alerts with the rider's GPS location in the event of a crash, notifying pre-registered contacts and the nearest hospital. By integrating multiple sensors and communication modules, the smart helmet addresses critical gaps in post-accident response and contributes to more effective emergency management.

II. LITERATURE REVIEW

Traditional accident prevention methods, such as manual helmet checks and traffic enforcement, are largely reactive. To overcome these limitations, Kumar *et al.* [1] developed a smart helmet using Arduino Uno for automatic accident detection and alerting. Later systems incorporated IoT-enabled platforms for helmet-wear monitoring to enhance compliance and reduce accident risk [2]. Moreover, alcohol detection sensors have been integrated to monitor rider intoxication and send alerts for timely intervention [3]. Combining these functions enables continuous monitoring and rapid assistance, improving overall rider safety.

III. MATERIALS AND METHODS

A. Hardware Setup

The system components include:

- **Microcontroller:** ESP32 as the central processing unit.
- **Alcohol Sensor:** MQ-3 sensor for detecting rider intoxication [3].
- **Helmet-Worn Sensor:** Reed switch module.
- **Crash Sensor:** MPU-6050 3-axis accelerometer for detecting crash events.
- **GPS Module:** NEO-6M GPS module (accuracy: 2.5 m) for real-time location tracking.

- **GSM Module:** SIM800L module for sending SMS-based emergency alerts.
- **Power Supply:** 3.7 V, 2000 mAh LiPo battery powers the system.

The complete hardware setup is illustrated in Fig. 1, which shows the interconnections between the ESP32, sensors, and communication modules.

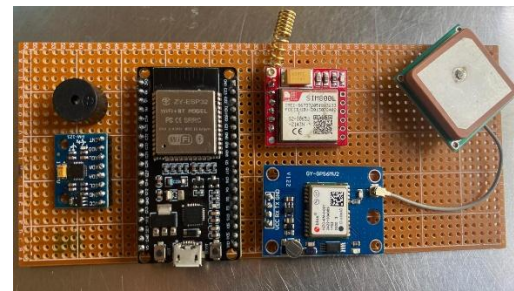


Fig. 1. Hardware prototype setup of the system.

B. System Architecture

All components are interfaced with the ESP32 microcontroller. The MQ-3 sensor connects via an analog input pin to measure alcohol levels. The reed switch monitors helmet usage to ensure other system functions operate only when the helmet is worn. The MPU-6050 connects via I2C to monitor acceleration in three axes. GPS and GSM modules communicate through serial pins. BLE provides near-instantaneous transmission of accident and location data to the mobile app.

C. Accident Detection

The MPU-6050 accelerometer measures acceleration along X, Y, and Z axes. Total acceleration A is calculated using:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (1)$$

where:

- A_x is the acceleration along the X-axis
- A_y is the acceleration along the Y-axis
- A_z is the acceleration along the Z-axis
- A represents the total resultant acceleration.

Equation (1) is used to distinguish between stable and accident conditions.

- **Stable Condition:** Acceleration $\approx 9.81 \text{ m/s}^2$ (1 g)

- Accident Condition: Acceleration > 117.7 m/s² (12 g), An emergency alert is triggered, with calibration performed using simulated crashes, such as dropping the helmet from controlled heights [3].

D. Alcohol Detection

The MQ-3 sensor provides an analog voltage output (0–3.3 V) proportional to ethanol concentration. The ESP32 reads this via its ADC using the formula:

$$\text{Voltage} = \text{sensor value} \times \frac{3.3}{4095} \quad (2)$$

where *sensor value* is the ADC reading in the range 0–4095. Equation (2) is used to convert the raw ADC values into the corresponding sensor voltage for further processing.

E. Helmet Usage Monitoring

A reed switch near the plug area aligns with a magnet on the opposite side. When the helmet is worn, the circuit closes; if open, the system detects the helmet is not in use.

F. Mobile App and BLE Integration

The helmet connects to a mobile app via BLE to display real-time alcohol levels. Upon a crash, the ESP32 sends GPS coordinates to the app, which identifies the nearest hospital using Google API.

IV. RESULTS AND DISCUSSION

Fig. 2 shows MQ-3 sensor readings over time, with ADC values (blue) and the 1800-threshold (red dashed), highlighting safe and elevated alcohol levels.

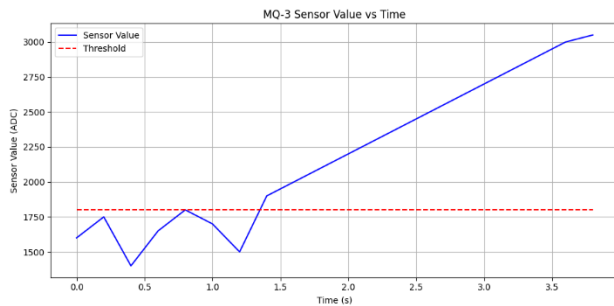


Fig. 2. Alcohol consumption level

Fig. 3 and Fig. 4 illustrate the system response during simulated crashes, with Fig. 3 showing stable conditions and Fig. 4 showing significant fluctuations during a crash.

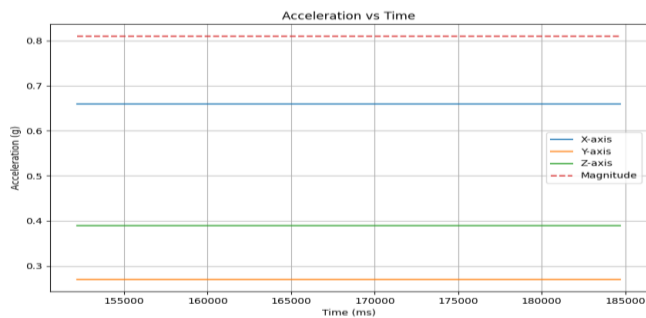


Fig. 3. Acceleration of the helmet is in a stable position.

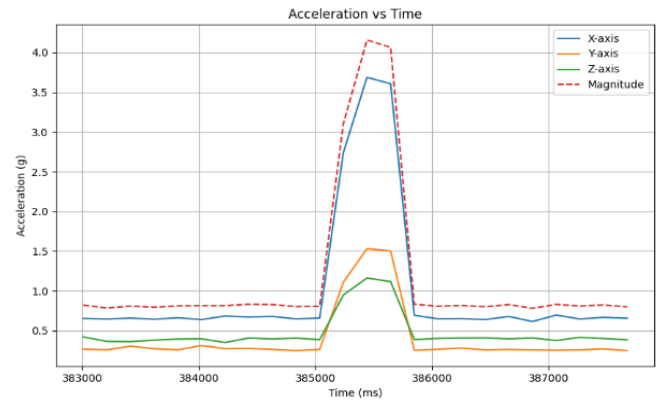


Fig. 4. Acceleration of the helmet during a simulated crash event

A. Performance Analysis

1. Rapid Detection – Continuously monitors helmet usage, alcohol levels, and crashes.
2. Instant Alerts – Sends notifications to pre-registered contacts and the nearest hospital for timely response.
3. Low-Power BLE Communication – Ensures efficient, reliable data transmission while preserving battery life.

B. Challenges and Limitations

1. Crash Detection – Setting a reliable MPU-6050 threshold was difficult; too low caused false alerts, too high missed real crashes.
2. GPS Accuracy – Location updates sometimes fluctuated, especially in urban areas.
3. App Integration – Maintaining smooth BLE communication required handling data loss and synchronization issues.

V. CONCLUSION

The Smart Helmet uses sensors and communication modules to monitor alcohol levels, helmet usage, and crashes. Alerts with GPS coordinates are sent via BLE and GSM to a mobile app. Despite calibration and GPS challenges, the system shows strong potential as a practical safety solution for motorcyclists.

ACKNOWLEDGMENT

I sincerely thank Dr. Sulochana Sooriyaarachchi for her guidance and acknowledge Weijith Wimal Siri and Yasantha Niroshana for their support.

REFERENCES

- [1] V. Kumar, A. MS, A. MH, D. CR, H. Gowda, "Smart helmet for accident detection using Arduino Uno," *Int. Res. J. Mod. Eng. Technol.Sci.*, vol.5, no.6, pp.1–5, 2023. Available: https://www.irjmets.com/uploadedfiles/paper/issue_6_june_2023/4_2705/final/fin_irjmets1688043322.pdf
- [2] J. Mathavan, V. K. D. Wijesekara, N. Satheeskant, W. M. U. J. Wanasinghe, M. Maathushan, V. V. Wijenayake, "Smart helmet to start the motorbike and to prevent accidents," *IOP Conf. Ser.: Mater. Sci. Eng.*, vol. 1258, no. 1, p. 012051, 2022. doi: 10.1088/1757-899X/1258/1/012051.
- [3] Mdpi, "Alcohol detection and monitoring for transportation safety," *MDPI Sensors*, 2023. Available: <https://www.mdpi.com/2673-4591/58/1/129>